

Multimodal Assistive Vision System

FOR THE VISUALLY IMPAIRED

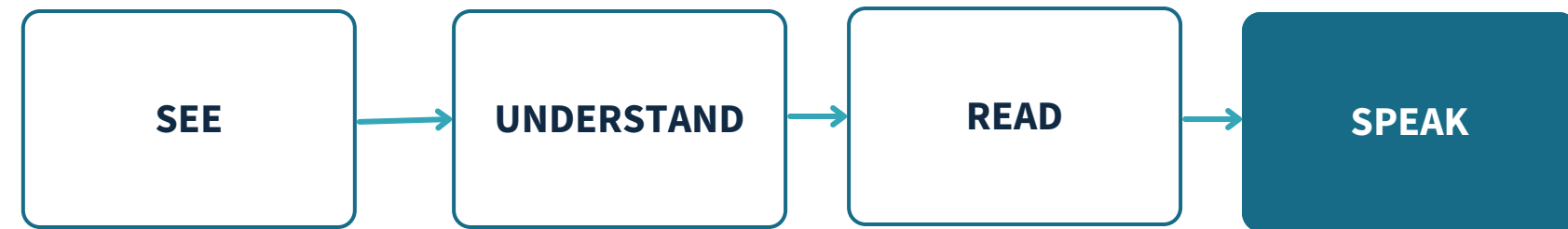
Using Spatial CNN and Scene Text Recognition

PHASE 1 REVIEW

50% implementation

Background & Motivation

- Visual impairment limits independent navigation
- Obstacles create everyday safety risks
- Signs provide essential environmental information
- Existing tools offer limited context
- Real-time assistance improves user independence



Problem Statement

“Existing assistive systems often handle obstacle detection and text reading separately, limiting complete environmental awareness.”

01 — SPATIAL AWARENESS

Where is the obstacle?

02 — TEXT RECOGNITION

What does the environment say?

03 — ACTIONABLE OUTPUT

What should the user hear?

04 — LOW LATENCY

Can it assist in real time?

Project Objectives

Two perception modules converge into one assistive output.



Literature Review

Related work indicates strong components—but incomplete integration.

STUDY	MAIN FOCUS	LIMITATION
Arystanbekov et al.	Edge VLM	High latency
Çaylı et al.	Spatial VLM	No dedicated OCR
Sudha et al.	Multi-task assistance	Hardware constraints
Islam et al.	Lightweight detection	Limited context
Gao et al.	OCR accessibility	No navigation

Speed, context and integrated assistance remain difficult to balance.

Research Gap

The unresolved gap lies between perception and useful assistance.

SINGLE-TASK SYSTEMS

Object detection OR OCR only

HIGH LATENCY

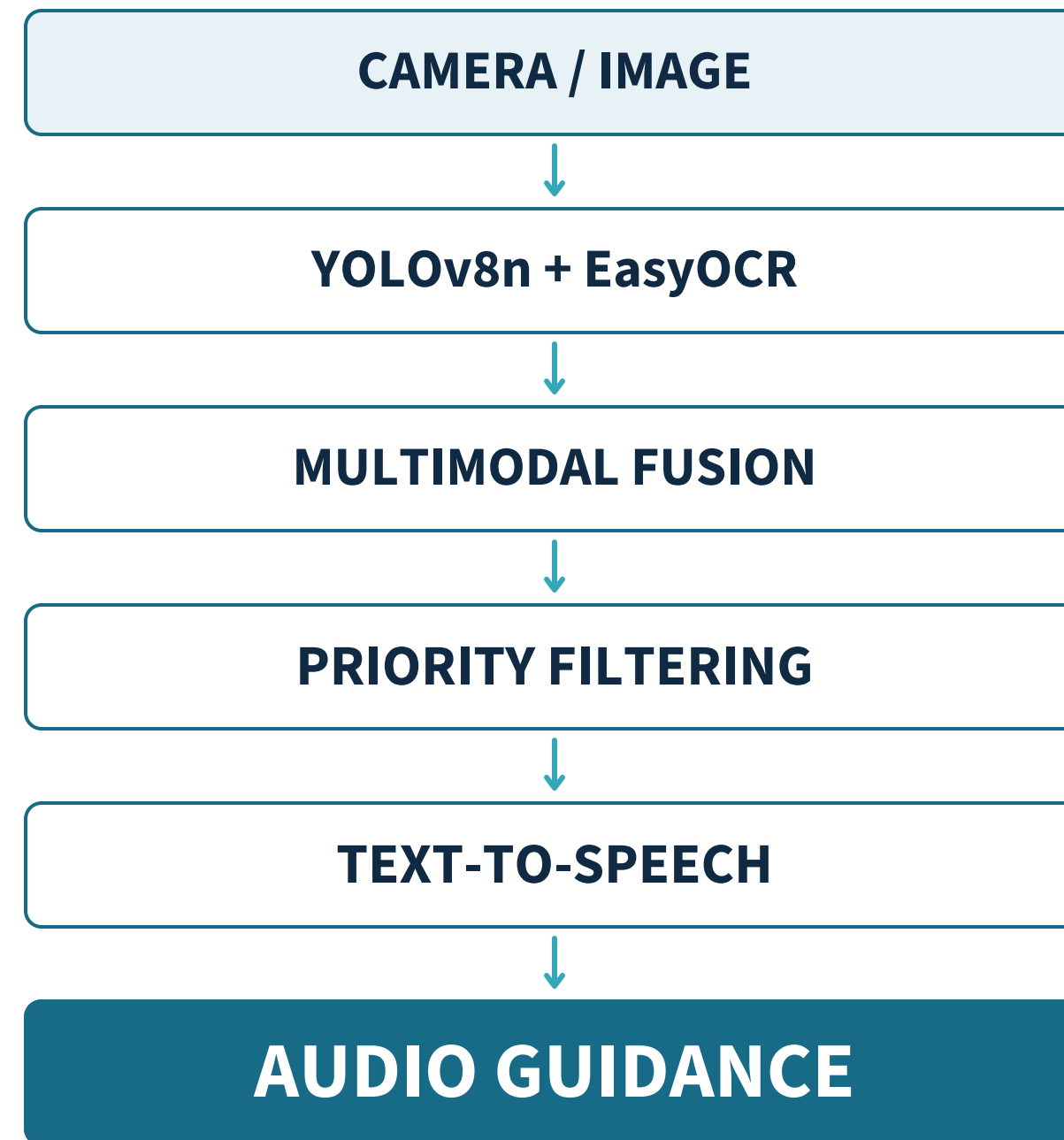
Heavy multimodal models
unsuitable for real-time

RAW OUTPUTS

Limited actionable guidance
from outputs

UNIFIED + LIGHTWEIGHT + ACTIONABLE

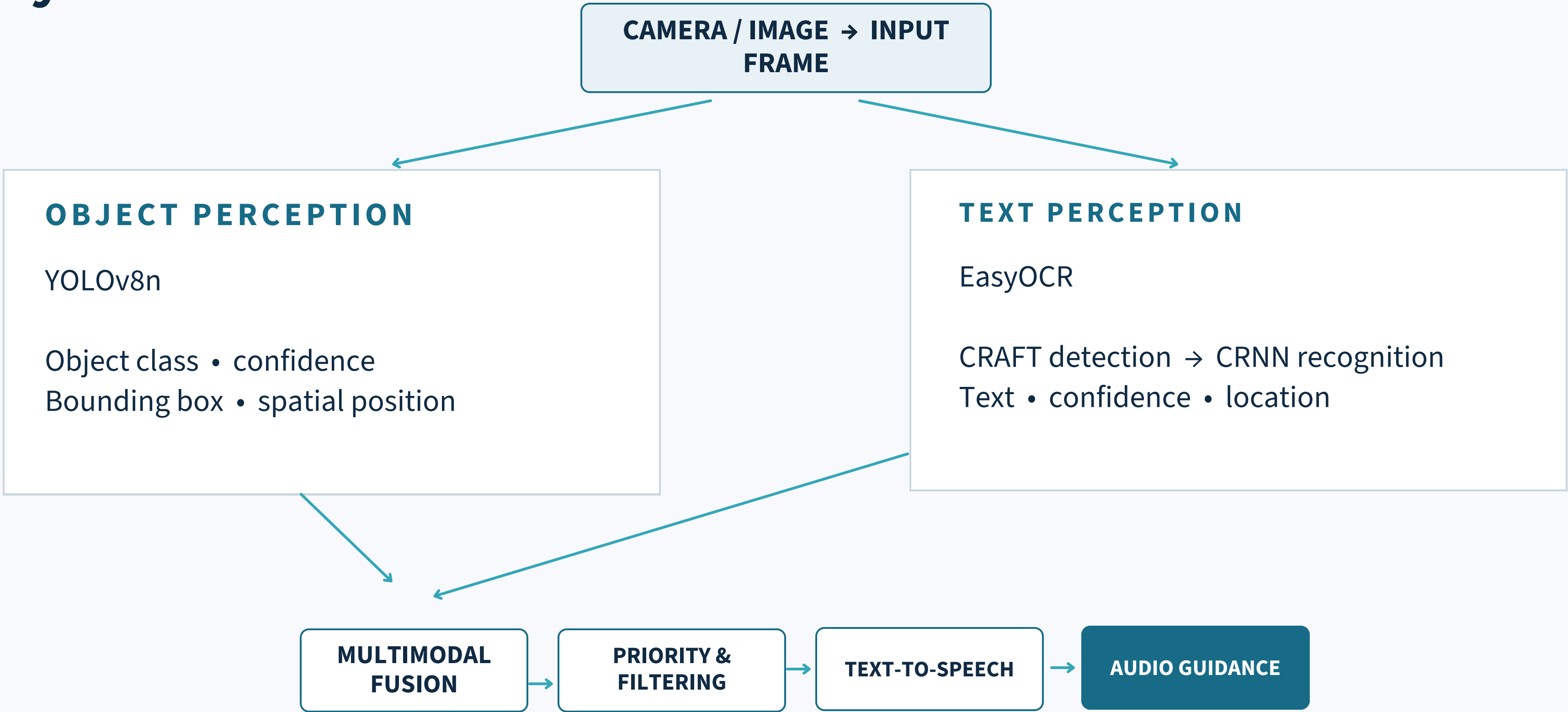
Proposed Solution



Convert visual surroundings into concise, actionable audio guidance.

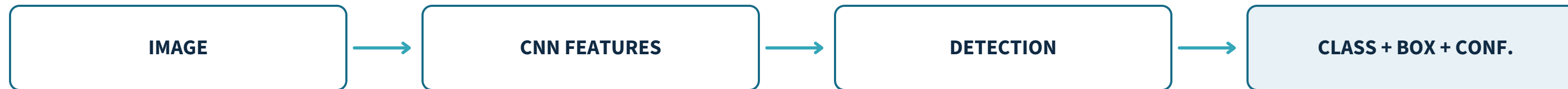
System Architecture

PHASE 1 / IMPLEMENTED SYSTEM

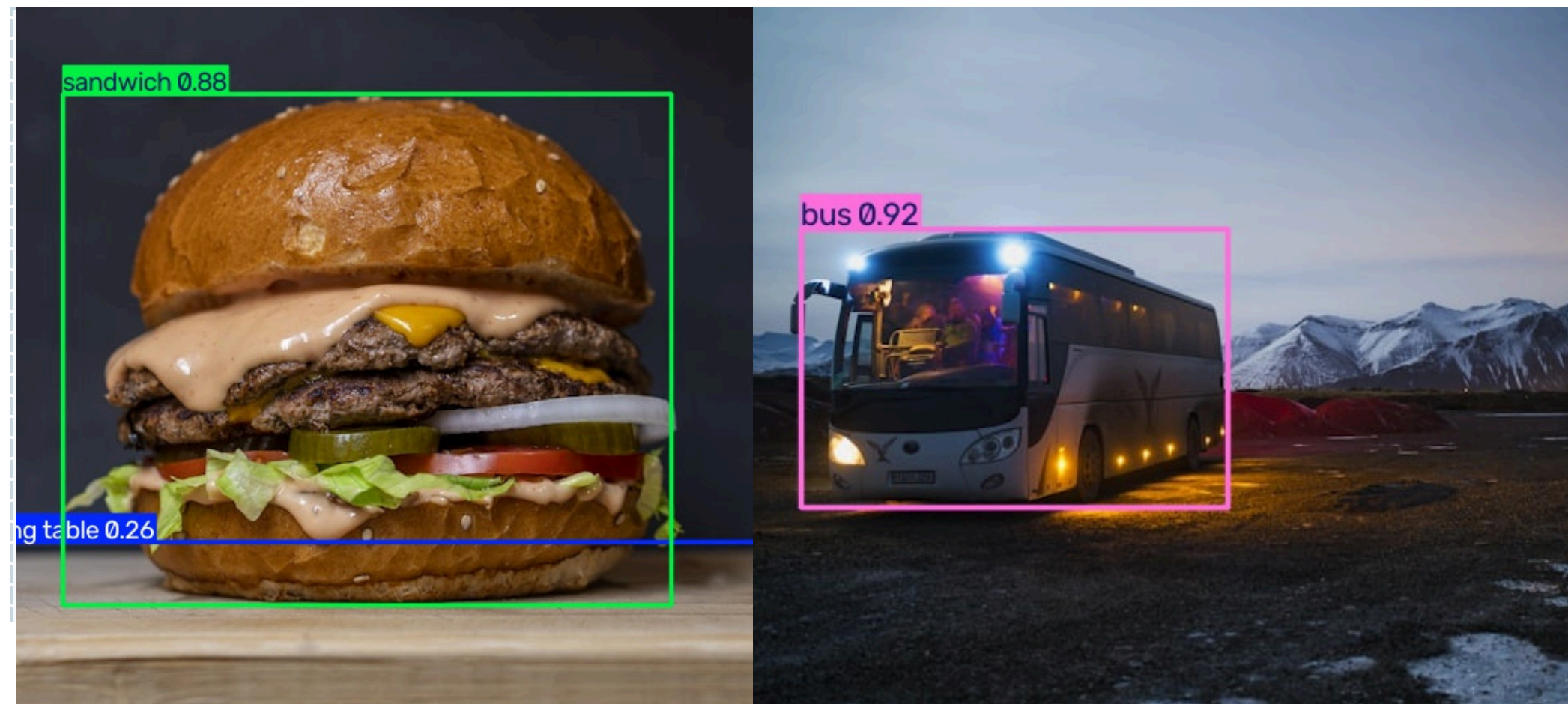


Module 1 — Object Detection

YOLOv8n

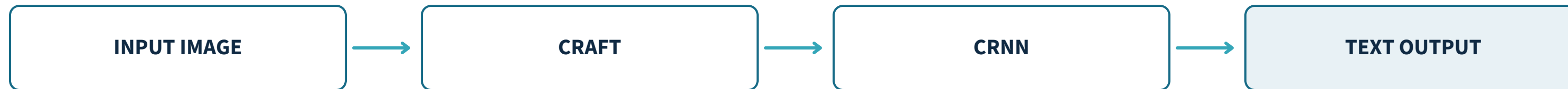


- Lightweight, single-stage detector
- Anchor-free architecture
- COCO pretrained model
- CPU-oriented implementation

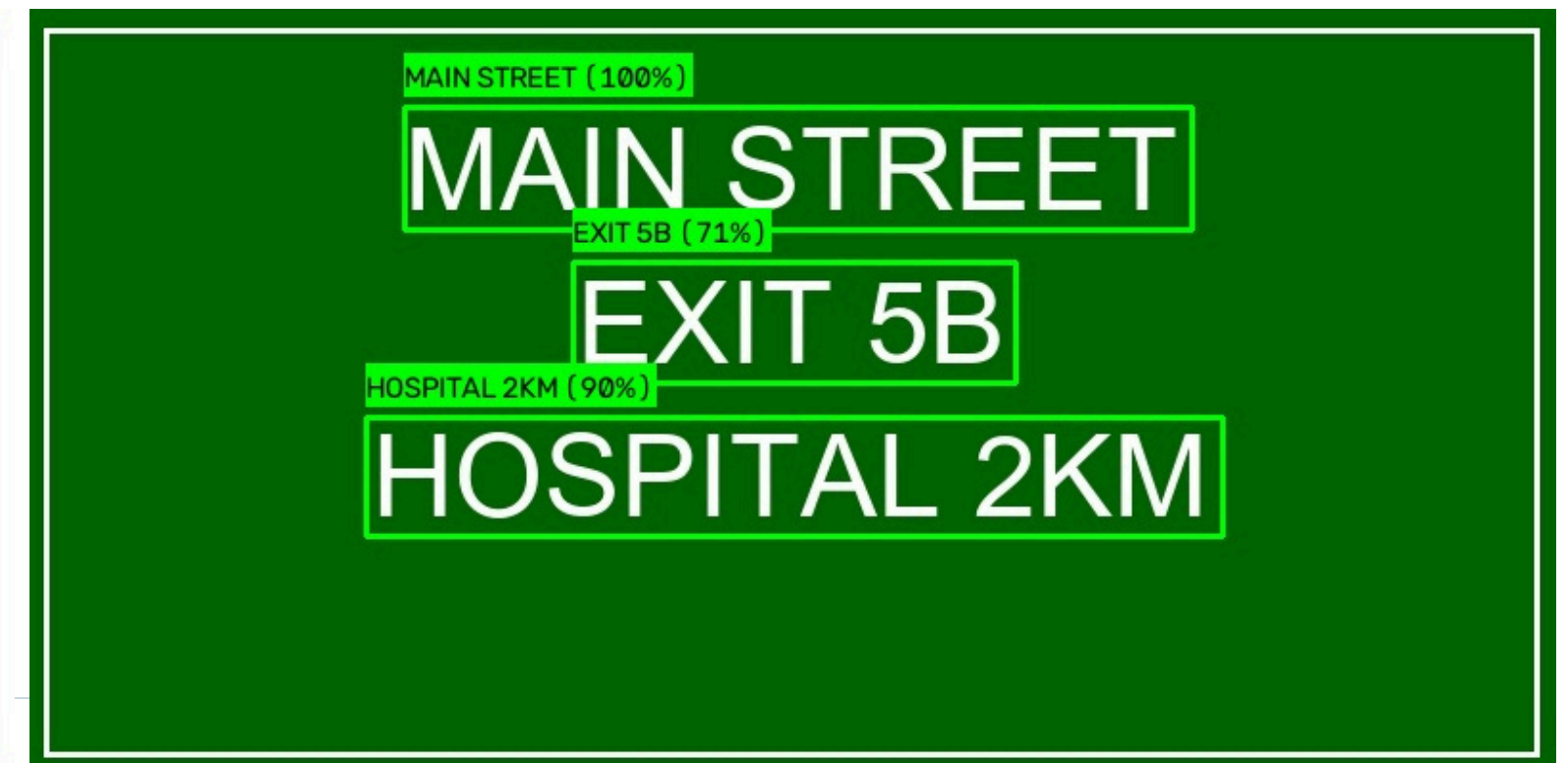
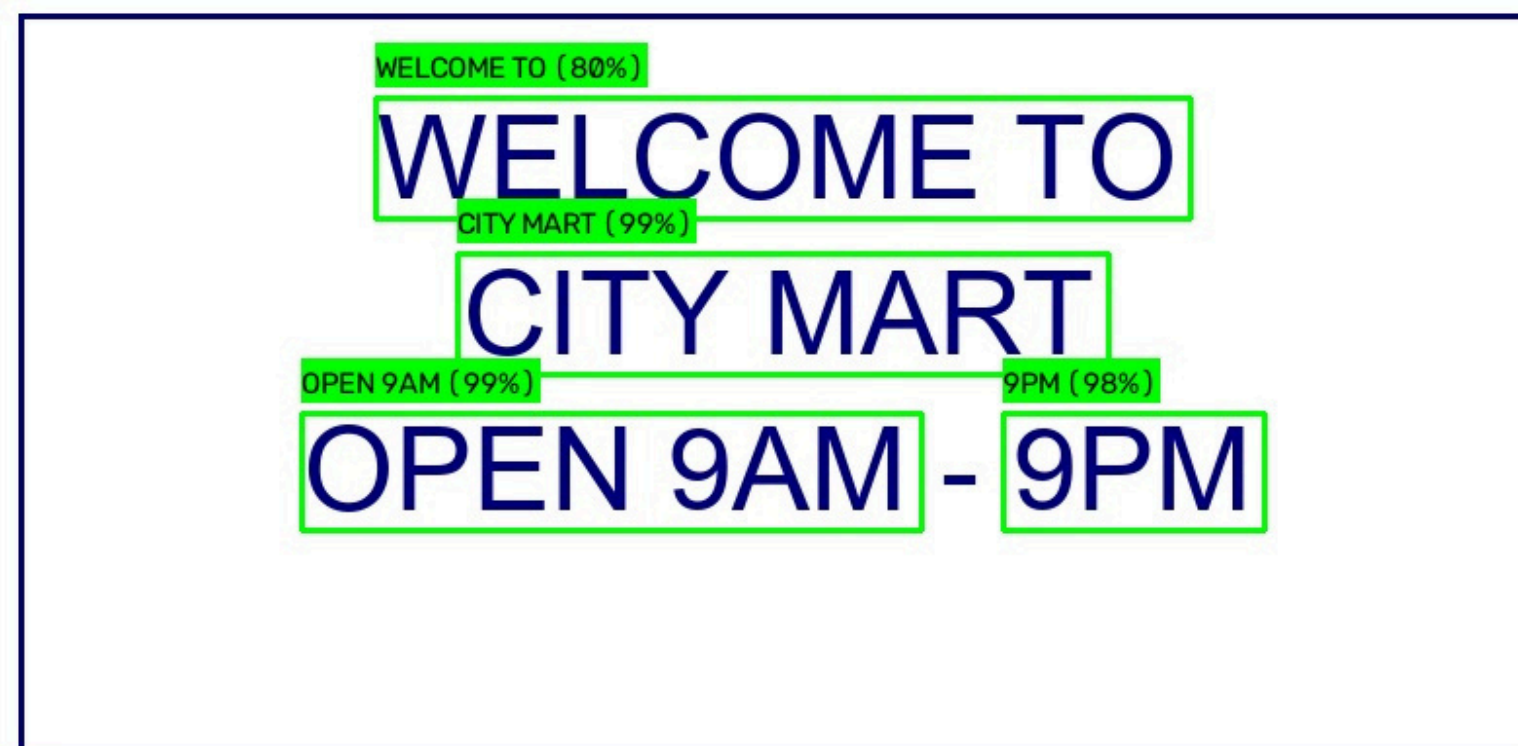


Module 2 — Scene Text Recognition

EasyOCR

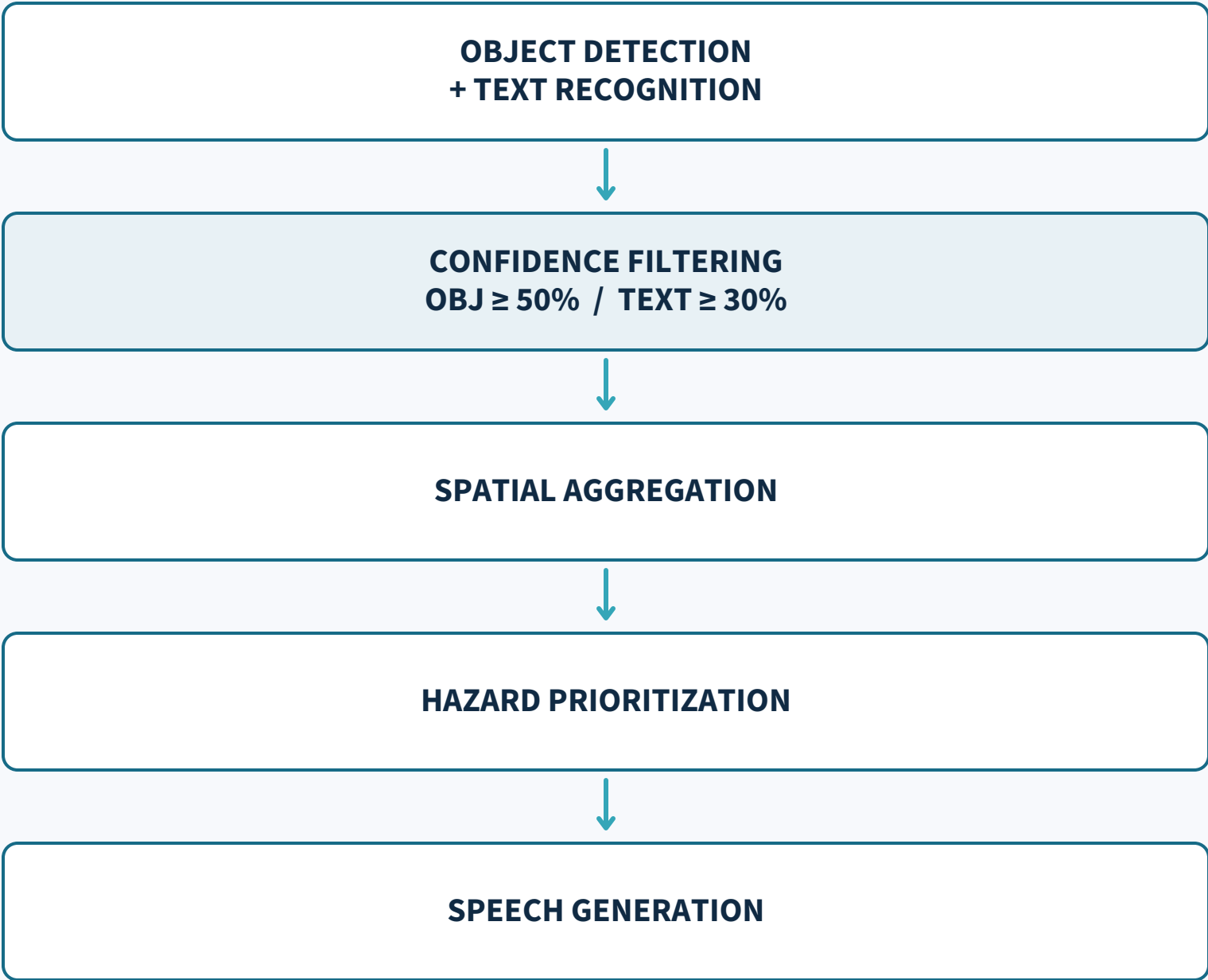


- CRAFT detects text regions
- CRNN recognizes character sequences
- BiLSTM captures sequence context
- CTC decodes variable-length text



Module 3 — Fusion & Speech

IMPLEMENTED THRESHOLDS + FUSION LOGIC



EXAMPLE SYSTEM OUTPUT

“Caution: 1 car and 1 person nearby. Text detected: Bus Stop.”

Dataset & Experimental Setup

TEST DATA

38 test images
Street scenes
Indoor scenes
Signs and text
Mixed / challenging scenes

TECHNOLOGY STACK

Python • PyTorch
Ultralytics • EasyOCR
OpenCV
CPU / Webcam

TEST MODES / Image-level testing + real-time webcam testing

Implementation Progress & Results

COMPLETED

- ✓ YOLOv8n pipeline
- ✓ EasyOCR integration
- ✓ Multimodal fusion
- ✓ Speech generation
- ✓ Image testing
- ✓ Live camera pipeline

PHASE 2 / PLANNED

- VizWiz fine-tuning
- Vision-Language Model
- Monocular depth estimation
- Edge / mobile deployment

PROJECT RESULTS / SCREENSHOTS
PLACEHOLDER

38

TEST IMAGES

150+

OBSTACLE DETECTIONS

50+

TEXT DETECTIONS

2–4 sec

PER IMAGE

Challenges & Phase 2 Plan

CURRENT CHALLENGES

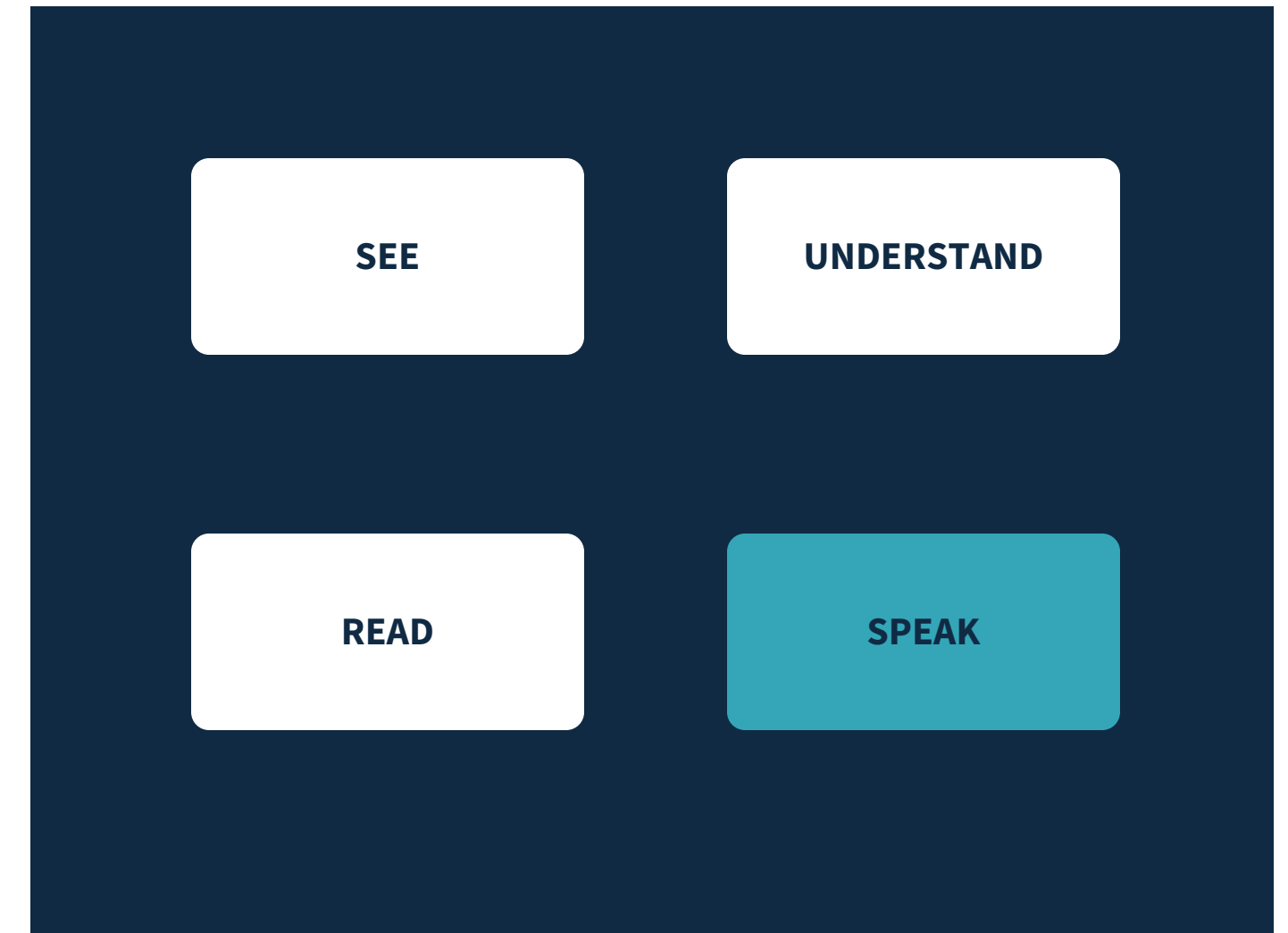
- CPU inference increases latency
- OCR struggles with poor-quality text
- Small text is difficult to recognize
- Audio requires prioritization
- Exact distance is unavailable

PHASE 2 / PLANNED

- 01 VizWiz fine-tuning
- 02 Vision-Language Model
- 03 MiDaS depth estimation
- 04 Edge / mobile deployment

Conclusion

Unified visual assistance pipeline developed
Object detection + scene text recognition integrated
Fusion generates concise audio guidance
End-to-end audio output demonstrated
Phase 2 roadmap established



SEE → UNDERSTAND → READ → SPEAK